

YUE FAN

Department of Electrical, Computer, and Systems Engineering

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EDUCATION

Case Western Reserve University, Cleveland, OH | Aug 2022 – Present

Ph.D. Candidate, Systems and Control Engineering (Expected May 2027)

Advisor: Behnam Malakooti, Ph.D.

Dissertation: Geometric Dispersion Theory for Decision Applications (tentative)

Oregon State University, Corvallis, OR | Mar 2021

B.S., Electrical & Computer Engineering; Minor in Computer Science

RESEARCH INTERESTS

Multi-criteria decision making (MCDM); Geometric Dispersion Theory (GDT); decision analytics and utility/value prediction; interpretable machine learning and hybrid ML–decision frameworks; decision making under risk; optimization modeling with applications in energy, finance, and quality assessment.

TECHNICAL SKILLS

Programming: Python (PyTorch, TensorFlow, scikit-learn, NumPy, pandas), MATLAB, C/C++

Optimization & Decision Science: convex optimization; linear, network, dynamic, integer, and nonlinear programming; MCDM (GDT, SAW, TOPSIS, VIKOR, AHP, MAUT, PROMETHEE); out-of-sample validation; experimental design

Machine Learning & Data Science: regression, classification, clustering, neural networks, SVM, feature selection, dimensionality reduction; distributed training (PyTorch Lightning, MPI)

Tools: Tableau, Excel Solver, LaTeX, Git; PCB design (KiCad)

RESEARCH IN PROGRESS

[1] Neural Networks with Geometric Dispersion Theory for Multi-Criteria Decision Making (Job Market Paper; preparing for submission to Management Science, expected October 2026)

Yue Fan, Huanshuo Wang, Behnam Malakooti

– Accepted for presentation at INFORMS Annual Meeting 2026, San Francisco

[2] Geometric Dispersion Utilities for Multi-Criteria Decision Making Under Criterion and Training-Sample Reduction (Under Review at IEEE Transactions on Systems, Man, and Cybernetics: Systems)

Yue Fan, Huanshuo Wang, Behnam Malakooti

[3] A Comparative Analysis of Multi-Criteria and Geometric Dispersion Theory Methods Using Out-of-Sample Validation with Application to Wind Farm Location Selection (In revision for resubmission)

Behnam Malakooti, Huanshuo Wang, Aziz [Last Name], Yue Fan

[4] Asymmetric GDT Distribution for Stock Portfolio Analysis (Working Paper)

Yue Fan, Huanshuo Wang, Behnam Malakooti

[5] GDT-Based Group Decision-Making Framework: Aggregating Preferences Across Multiple Decision Makers (Early Stage)

Yue Fan, Behnam Malakooti

[6] Geometric Dispersion Theory as an Interpretable Alternative to SHAP (Early Stage)

Huanshuo Wang, Yue Fan

PUBLICATIONS AND PATENTS

Fan, Y., & Wang, J. (2022). Solar cells fabrication procedures. In Proceedings of the 3rd International Conference on Electronic Science and Automation Control (ESAC 2022). Bangkok, Thailand: Clausius Scientific Press.
<https://doi.org/10.23977/esac2022.004>

Fan, Y. (2021). Modular pet feeding device. Chinese Utility Model Patent No. ZL 2021 2 2413328.4 (filed October 2021; granted).

RESEARCH EXPERIENCE

Network Lab, Case Western Reserve University | Sep 2023 – Present

Graduate Research Assistant (Advisor: Behnam Malakooti, Ph.D.)

Established the mathematical foundations of Geometric Dispersion Theory (GDT), proving that ranking-only MCDM methods (TOPSIS, VIKOR) cannot recover absolute utility values because their scores shift with the alternative set, while GDT produces context-independent utilities

Built hybrid neural-network–GDT models that preserve weight-normalization constraints during training, retaining interpretable criterion weights while matching black-box predictive accuracy; validated across four domains (wind-farm siting, Amazon product ratings, health-insurance claims, portfolio returns), improving out-of-sample error by 23% over standard MCDM baselines.

Benchmark interpretable GDT utilities against traditional MCDM and predictive models for accuracy, stability, and interpretability under criterion and training-sample reduction (red-wine quality, social-worker decisions, and housing-price data).

Semiconductor Device Physics and Fabrication Seminar | Oct 2021 – Dec 2021

Research Team Lead (Supervisor: Munib Wober, Ph.D.)

Led a 5-person team evaluating wafer-cleaning and doping sequences for solar-cell efficiency; designed a P+/N-type cell structure that became the basis of the team's ESAC 2022 conference paper.

TEACHING EXPERIENCE

Teaching Assistant with Lecturing and Course Development, Case Western Reserve University

ECSE 360/460 Manufacturing and Automated Systems: Design and Optimization | Fall 2025

Co-designed the Multi-Criteria Decision-Making module and delivered lectures

ECSE 352 Engineering Economics and Decision Analysis | Fall 2024, Fall 2025

Designed the Risk Decision module (mean–variance analysis, decision trees, expected utility theory, GDT) and taught multiple lectures

ECSE 350/450 Operations and Systems Design | Spring 2026

Led Excel Solver instruction and exam review sessions; graded case-study projects

Teaching Assistant, Case Western Reserve University

ECSE 416 Convex Optimization in Engineering | Spring 2024, Spring 2025

ECSE 346 Engineering Optimization | Spring 2024, Spring 2025

EPOM 405 / ECSE 411 Applied Engineering Statistics (online graduate course) | Spring 2026

ECSE 408 Introduction to Linear Systems | Fall 2023

BUAI 445 Advanced Marketing Analytics (Weatherhead School) | Spring 2023

PROFESSIONAL EXPERIENCE

Olababy, Beaverton, OR | Jun 2021 – Mar 2022

Product Development Manager

Led development of an automatic baby-formula dispenser from technical requirements and feasibility analysis through PCB design and an engineering validation prototype.

Translated brand and market requirements into technical specifications as liaison among design, engineering, and marketing teams; supported supply-chain operations.

SELECTED TECHNICAL PROJECTS

Distributed Model Training with PyTorch Lightning and MPI | Fall 2023

Implemented distributed data-parallel training across multiple GPUs (BERT sentiment classification on IMDb); fine-tuned GPT-2 under constrained compute with real-time resource monitoring and early stopping.

PET Image Reconstruction Using Convex Optimization | Spring 2024

Formulated PET reconstruction as log-likelihood maximization; implemented gradient-ascent and quasi-Newton methods exploiting sparsity (1,080 voxels, 2,164 detector pairs); analyzed MLE vs. penalized-likelihood trade-offs.

Advanced Statistical Learning Implementations | Fall 2024

Built custom implementations of clustering, LAD/percentile regression, LASSO/ridge shrinkage, SVM, kNN, and multidimensional scaling; benchmarked against standard libraries.

SERVICE AND PROFESSIONAL AFFILIATIONS

Peer Reviewer, INFORMS, 2026

IEEE Student Member, since 2024

XLab Member, Case Western Reserve University, since 2023

REFERENCES

Behnam Malakooti, Ph.D.

Professor, Department of Electrical, Computer, and Systems Engineering, Case Western Reserve University

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Vira Chankong, Ph.D.

Associate Professor, Department of Electrical, Computer, and Systems Engineering, Case Western Reserve University

vira.chankong@case.edu